CSC1058 USER INTERFACE DESIGN & IMPLEMENTATION

MODULE COORDINATOR & LECTURER
Hyowon Lee

TEACHING ASSISTANT
Haoyan Zhang

	Tuesday (3 – 4pm)	Wednesday (11am - 12pm)
Week 1 (7 Sep)	Introduction/ orientation	HCI background
Week 2 (14 Sep)	Design guidelines	Design guidelines
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Week 4 (28 Sep)	Design guidelines	Project brainstorming
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Week 12 (23 Nov)	Summary/ reflection	Summary/ reflection

DCU

1

Guideline 2: Simple and natural dialogue

- Minimise the *artefacts*
- Grouping – Separating - Prioritising - Sharpening
- Readable fonts

What is the best font for my UI?

DCU

2

Serif family

Usability is very important … Times New Roman

Usability is very important … Cambria

Usability is very important … Georgia

Usability is very important … Century

Usability is very important … Palatino

⋮

3

Sans-Serif family

Usability is very important … Arial

Usability is very important … Calibri

Usability is very important … Helvetica

Usability is very important … Tribuchet MS

Usability is very important … Century Gothic

⋮

4

Usability is very important … Times New Roman (Serif)

Usability is very important … Arial (Sans-Serif)

USABILITY

USABILITY

5

Mr. and Mrs. Dursley, of number four, Privet Drive, were proud to say that they were perfectly normal, thank you very much. They were the last people you'd expect to be involved in anything strange or mysterious, because they just didn't hold with such nonsense.

Mr. Dursley was the director of a firm called Grunnings, which made drills. He was a big, beefy man with hardly any neck, although he did have a very large mustache. Mrs. Dursley was thin and blonde and had nearly twice the usual amount of neck, which came in very useful as she spent so much of her time craning over garden fences, spying on the neighbors. The Dursleys had a small son called Dudley and in their opinion there was no finer boy anywhere.

The Dursleys had everything they wanted, but they also had a secret, and their greatest fear was that somebody would discover it. They didn't think they could bear it if anyone found out about the Potters. Mrs. Potter was Mrs. Dursley's sister, but they hadn't met for several years; in fact, Mrs. Dursley pretended she didn't have a sister, because her sister and her good-for-nothing husband were as

from Harry Potter and the Sorcerer's Stone, Chapter I

Times New Roman (Serif)

6

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2:
ue
from Harry Potter and the Sorcerer's Ston

Making sans-serif for lengthy text slows down the reading

Arial (Sans-Serif)

7

MR. AND MRS. DURSLEY, OF NUMBER FOUR, PRIVET DRIVE, WERE PROUD TO SAY THAT THEY WERE PERFECTLY NORMAL, THANK YOU VERY MUCH. THEY WERE THE LAST PEOPLE YOU'D EXPECT TO BE INVOLVED IN ANYTHING STRANGE OR MYSTERIOUS, BECAUSE THEY JUST DIDN'T HOLD WITH SUCH NONSENSE.

MR. DURSLEY WAS THE DIRECTOR OF A FIRM CALLED GRUNNINGS, WHICH MADE DRILLS. HE WAS A BIG, BEEFY MAN WITH HARDLY ANY NECK, ALTHOUGH HE DID HAVE A VERY LARGE MUSTACHE. MRS. DURSLEY WAS THIN AND BLONDE AND HAD NEARLY TWICE THE USUAL AMOUNT OF NECK, WHICH CAME IN VERY USEFUL AS SHE SPE
MUCH OF HER TIME CRANING OVER GARDEN FENCES, SPYING
NEIGHBORS. THE DURSLEYS HAD A SMALL SON CALLED DUDL
THEIR OPINION THERE WAS NO FINER BOY ANYWHERE.

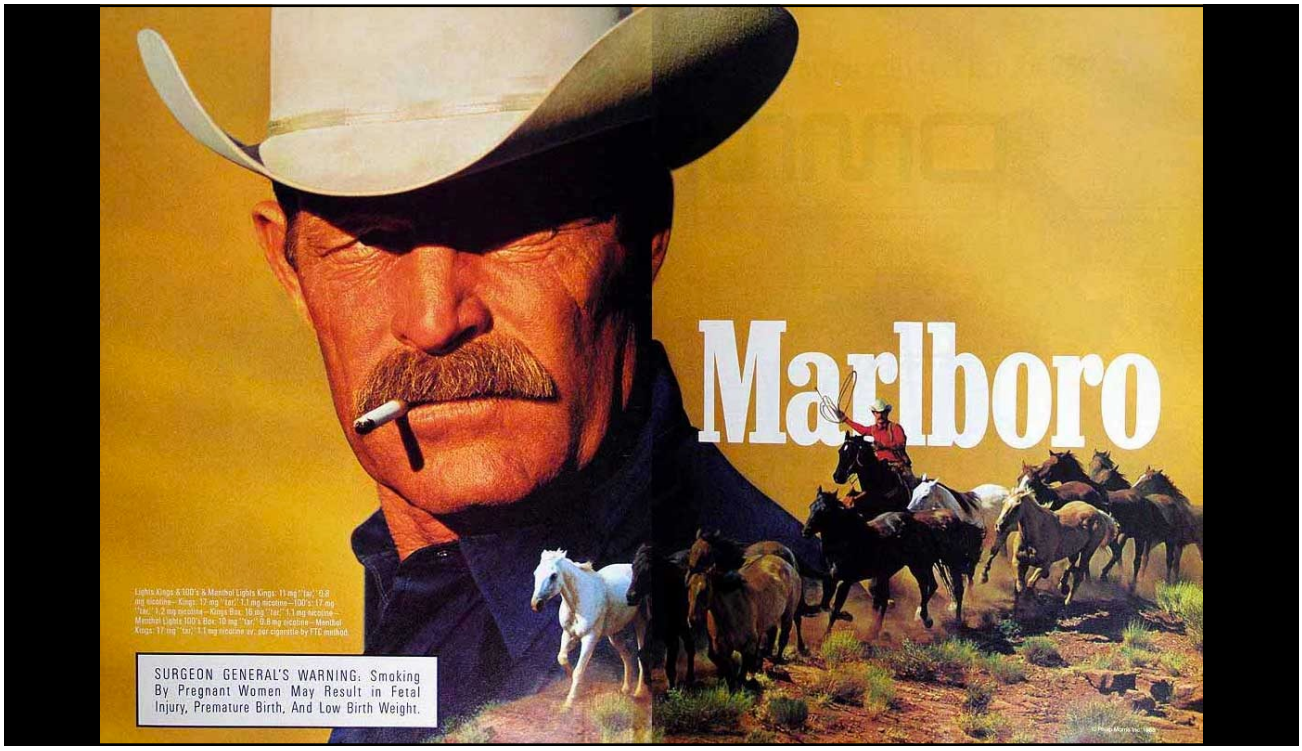
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HAD A SECRET, AND THEIR GREATEST FEAR WAS THAT SOMEBODY WOULD DISCOVER IT. THEY DIDN'T THINK THEY COULD BEAR IT IF ANYONE FOUND OUT ABOUT THE POTTERS. MRS. POTTER WAS MRS.

2:
ue
from Harry Potter and the Sorcerer'

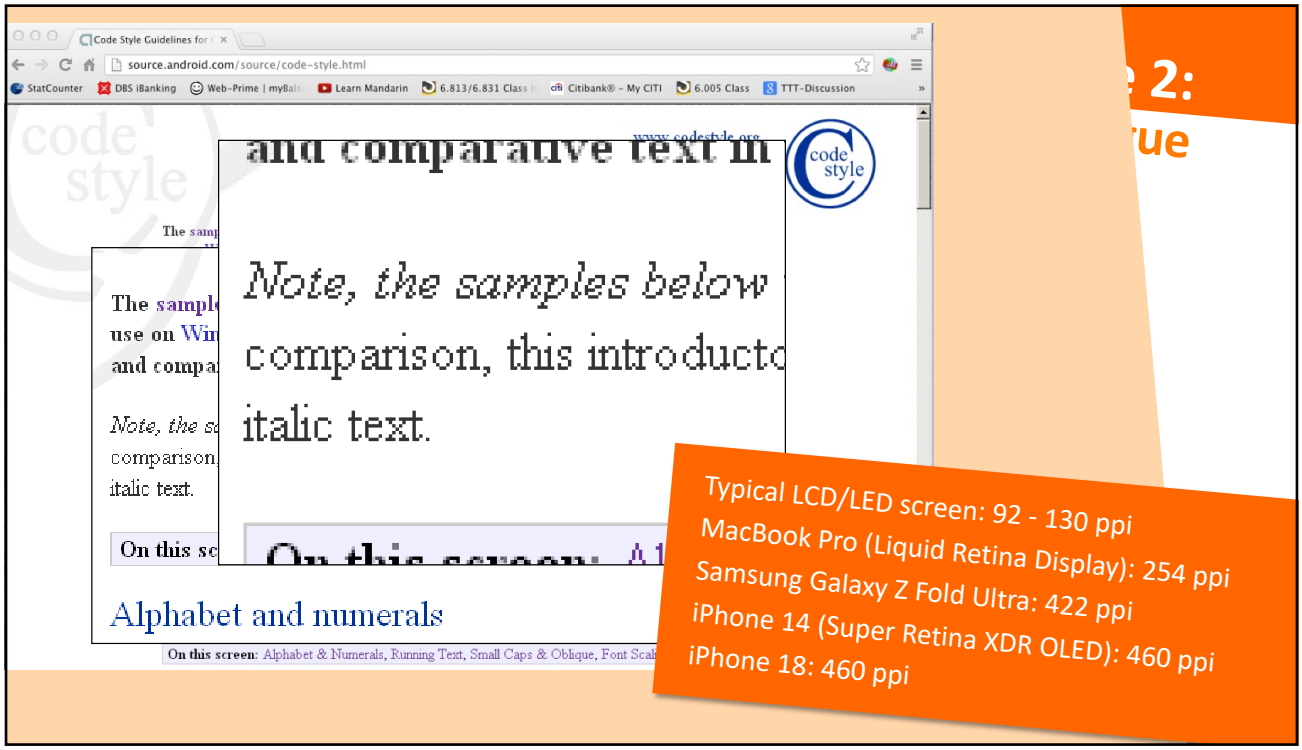
Making sans-serif + capitalization for lengthy text makes reading extremely difficult, considerably slowing down the reading (think about why)

Arial (Sans-Serif) All Caps

8


Marlboro
SURGEON GENERAL'S WARNING: Smoking By Pregnant Women May Result in Fetal Injury, Premature Birth, And Low Birth Weight.

9


Code Style Guidelines for
source.android.com/source/code-style.html
StatCounter
DBS iBanking
Web-Prime | myBa
Learn Mandarin
6.813/6.831 Class
Citibank® - My CITI
6.005 Class
TTT-Discussion
code style
www.codestyle.org
and comparative text in
Note, the samples below
comparison, this introducto
italic text.
The sample
use on Win
and compa
Note, the sa
comparison
italic text.
On this sc
Alphabet and numerals
On this screen: Alphabet & Numerals, Running Text, Small Caps & Oblique, Font Scal
2:
ue
Typical LCD/LED screen: 92 - 130 ppi
MacBook Pro (Liquid Retina Display): 254 ppi
Samsung Galaxy Z Fold Ultra: 422 ppi
iPhone 14 (Super Retina XDR OLED): 460 ppi
iPhone 18: 460 ppi

10

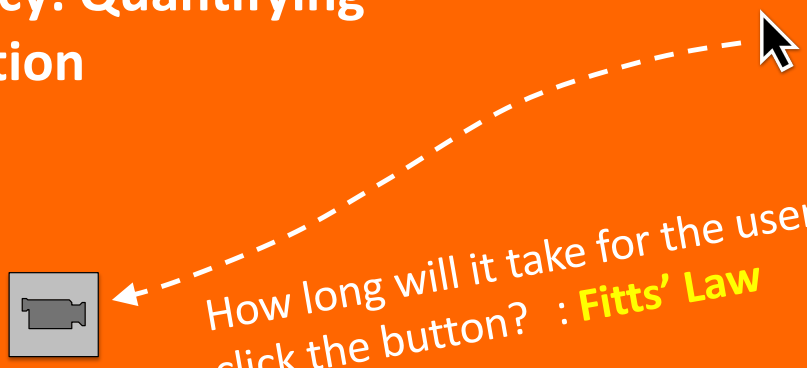
Guideline 2: Simple and natural dialogue

- Minimise the *artefacts*
- Grouping – Separating - Prioritising - Sharpening
- Readable fonts

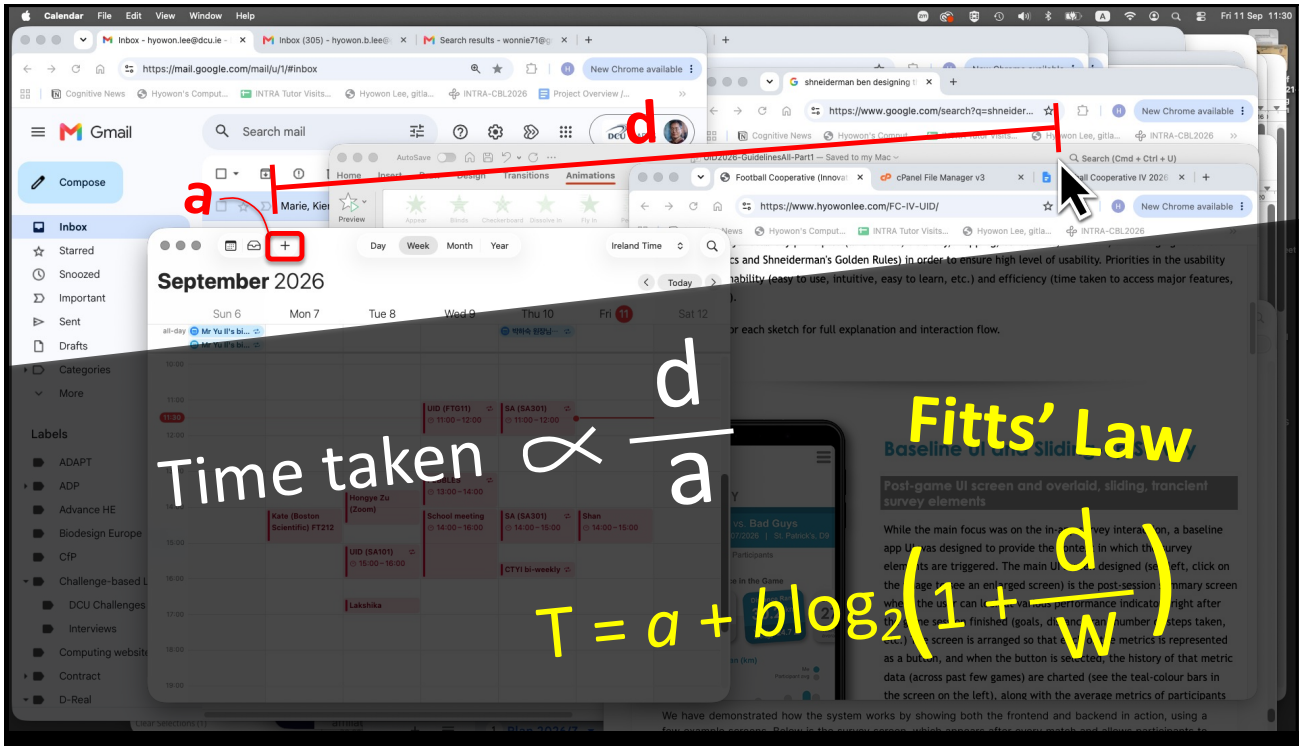
DCU

11

Efficiency: Quantifying Interaction



12



13

ACTIVITY (8 min)

On your favourite desktop/laptop software, identify how Fitt's Law can be used to reduce the overall interaction time in conducting a task

1. Write down 1 sentence that summarises the issue
2. Sketch the UI that reduces the interaction time

14

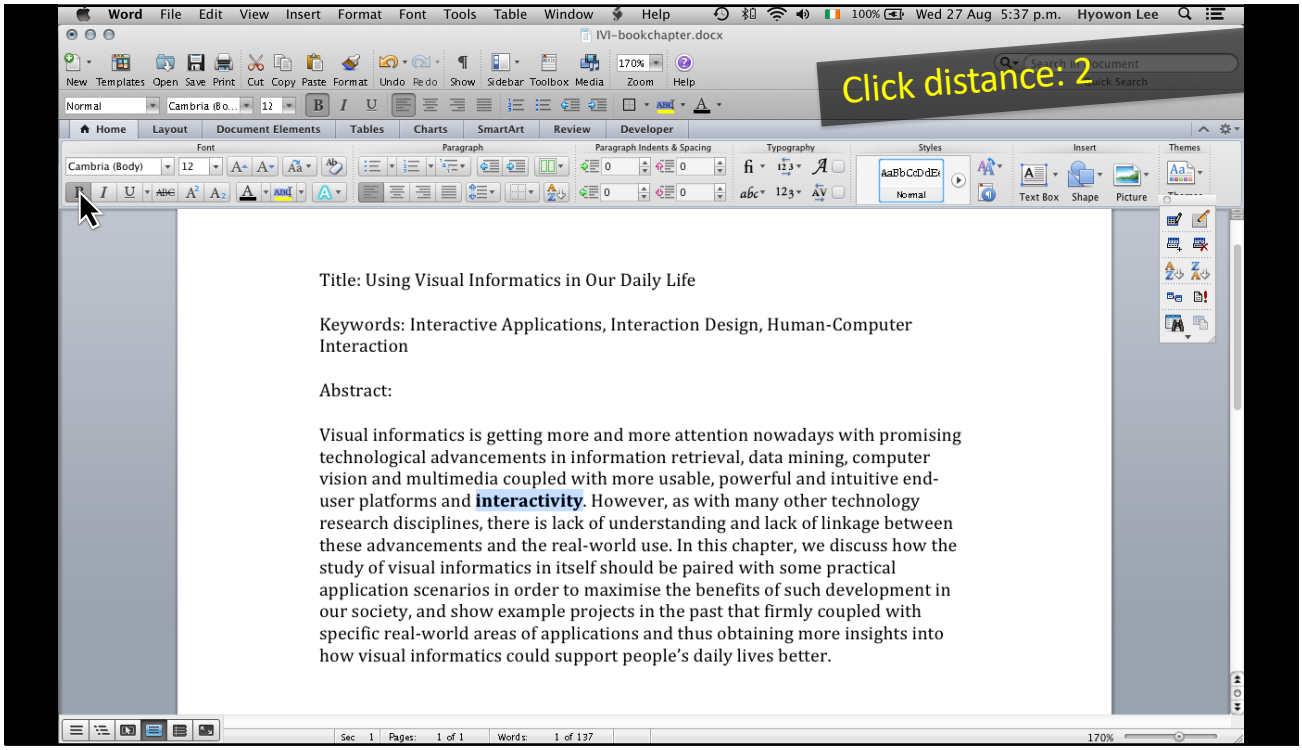
Efficiency: Quantifying Interaction



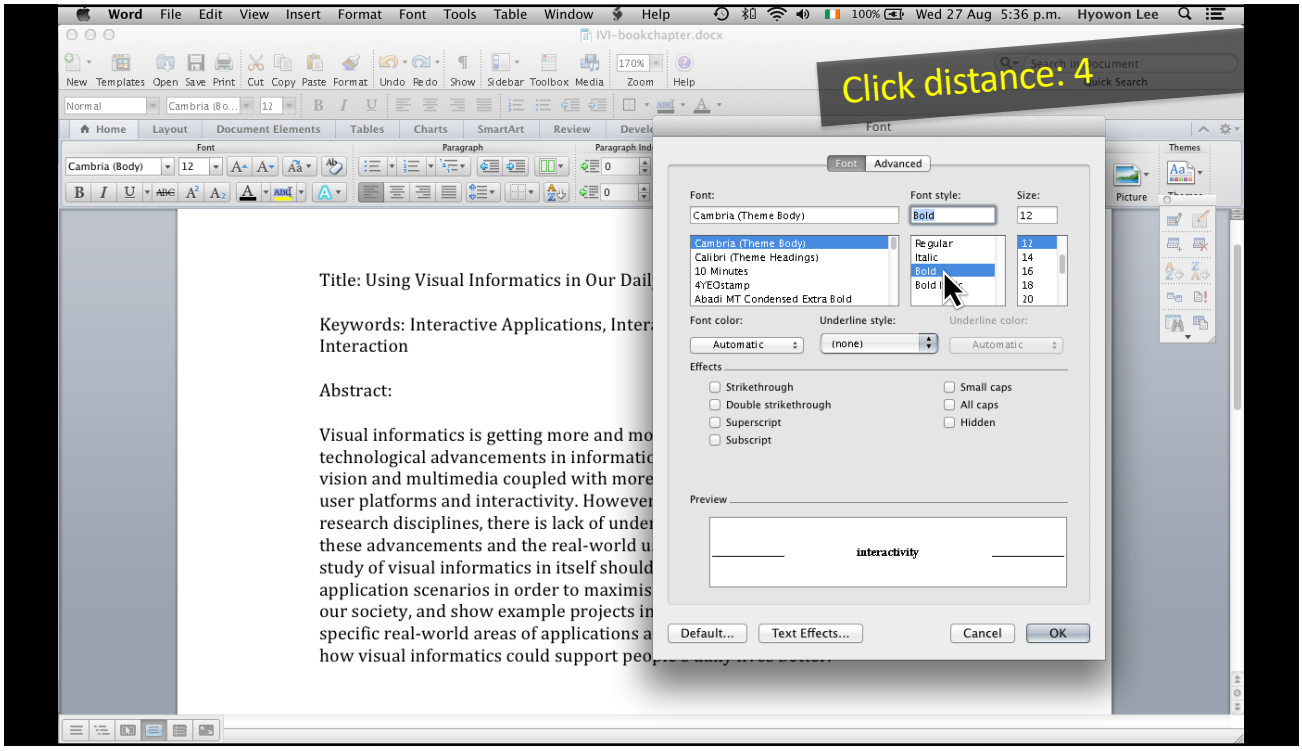
How long will it take for the user to click the button? : **Fitts' Law**

… How many mouse clicks are required to complete a task? : **Click distance**

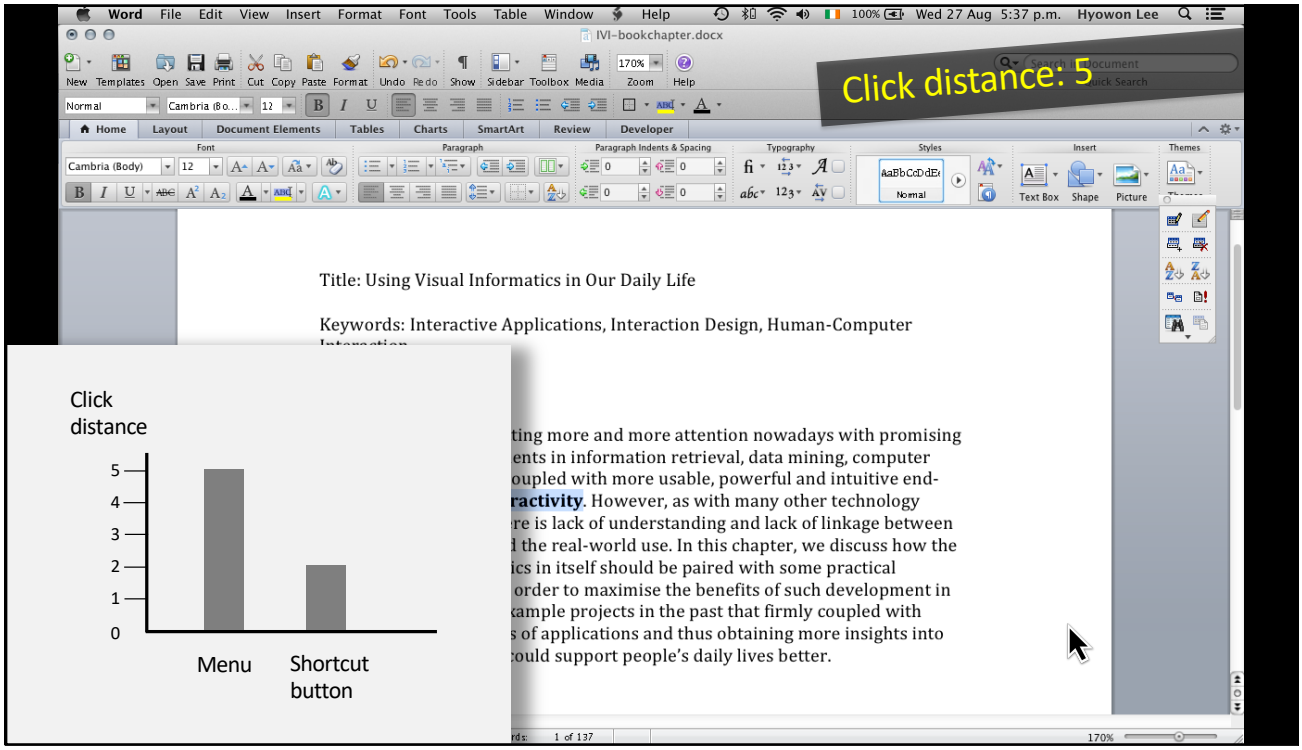
15



16


Click distance: 4
Font
Font
Advanced
Font:
Cambria (Theme Body)
Font style:
Bold
Size:
12
Regular
Italic
Bold
Bold I
14
16
18
20
Font color:
Automatic
Underline style:
(none)
Underline color:
Automatic
Effects
Strikethrough
Double strikethrough
Superscript
Subscript
Small caps
All caps
Hidden
Preview
interactivity
Default...
Text Effects...
Cancel
OK
Title: Using Visual Informatics in Our Dail
Keywords: Interactive Applications, Inter
Interaction
Abstract:
Visual informatics is getting more and mo
technological advancements in informatic
vision and multimedia coupled with more
user platforms and interactivity. However
research disciplines, there is lack of under
these advancements and the real-world u
study of visual informatics in itself should
application scenarios in order to maximis
our society, and show example projects in
specific real-world areas of applications a
how visual informatics could support peo

17


Click distance: 5
Title: Using Visual Informatics in Our Daily Life
Keywords: Interactive Applications, Interaction Design, Human-Computer
ting more and more attention nowadays with promising
ents in information retrieval, data mining, computer
oupled with more usable, powerful and intuitive end-
ractivity. However, as with many other technology
re is lack of understanding and lack of linkage between
l the real-world use. In this chapter, we discuss how the
ics in itself should be paired with some practical
order to maximise the benefits of such development in
xample projects in the past that firmly coupled with
s of applications and thus obtaining more insights into
ould support people's daily lives better.
Click distance
5
4
3
2
1
0
Menu
Shortcut button

18

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DCU

19

Efficiency: Quantifying Interaction

5 Usability Criteria
- Efficiency
- Learnability
- Memorability
- Error rate
- Satisfaction

How long will it take for the user to click the button? : **Fitts' Law**

ouse clicks are
plete a task? : **Click distance**

20

Efficiency: Quantifying Interaction

5 Usability Criteria

- Efficiency
- Learnability
- Memorability
- Error rate
- Satisfaction

Do **Fitts' Law** and **Click distance** enhance usability?

21

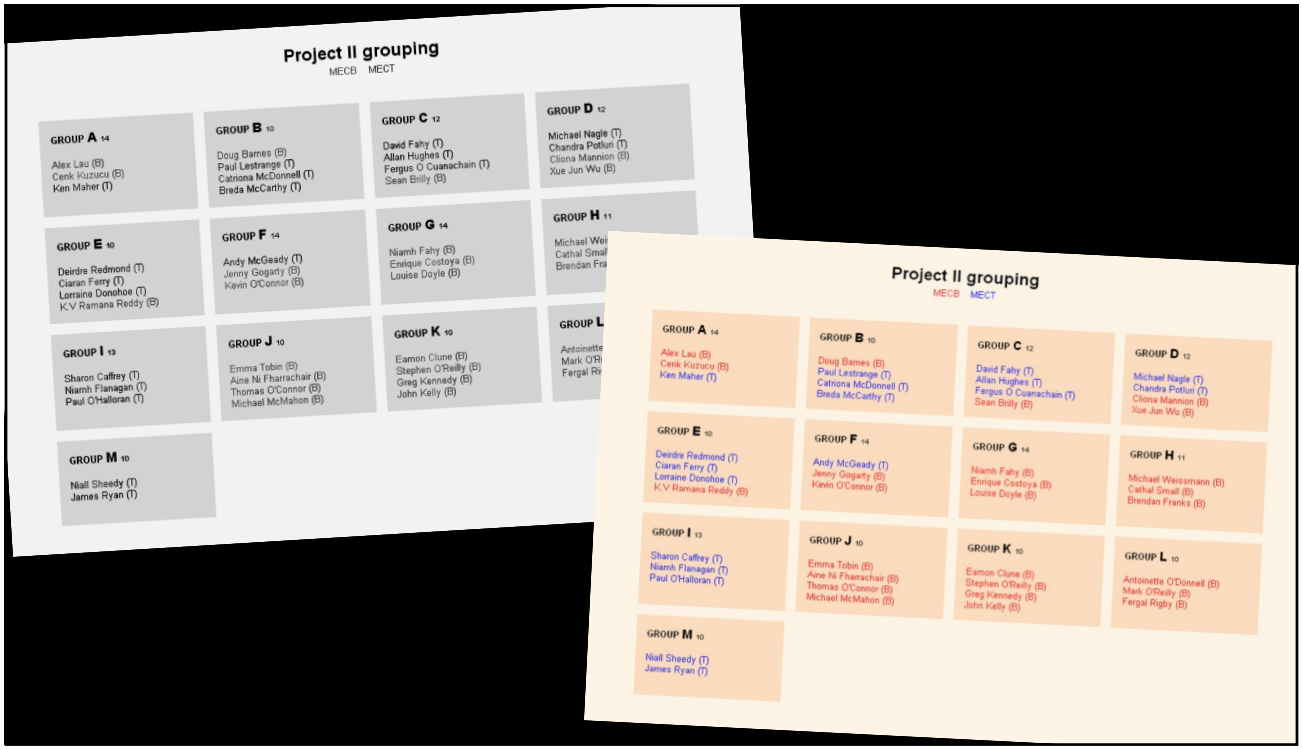
Guideline 2: Simple and natural dialogue

- Minimise the *artefacts*
- Grouping – Separating - Prioritising - Sharpening
- Readable fonts
- Good use of colour

22

- Use conservatively
- Use it for sharpening, prioritising, grouping, etc…
- Be careful with unintended effects
- Colour-coding

23



24

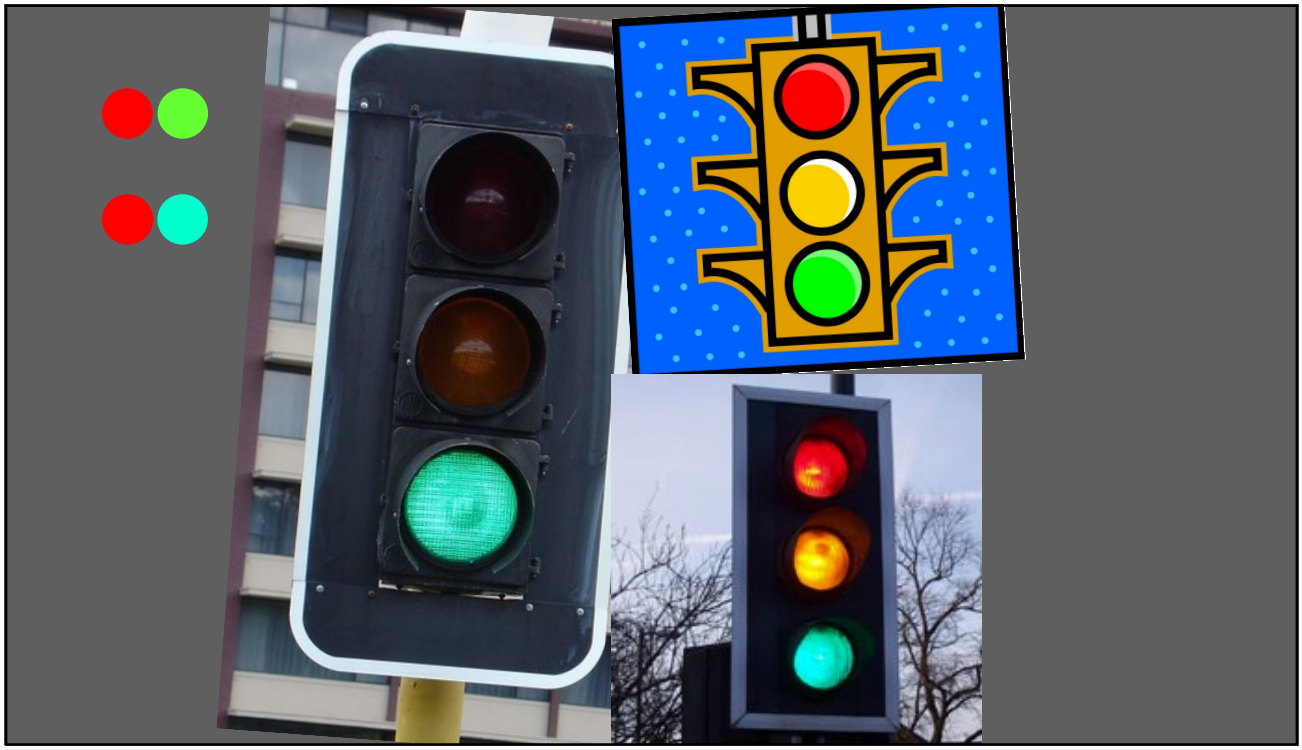
Project II grouping

MECT MECB

GROUP A	GROUP B	GROUP C	GROUP D
Robert Brennan (T) Shumin Cao (T) Conor Doyle (B) Roy Johansen (B)	Charles Fryer (T) Chi Luo (T) Francesca Feeney (B) Ultan O'Broin (B)	Kevin Costello (T) Teng Yu Wang (T) Louis Kelleher (B) Wojciech Wrona (B)	Bryan O'Sullivan (T) Mussa Elsaeid (T) Ronan Fitzsimons (B) Na Xing (B)
GROUP E	**GROUP F**	**GROUP G**	**GROUP H**
Ronan Flannery (T) Hao Ye (T) Andrew Osborne (B) Pradeep Pothakamuri (B)	Richard Howlett (T) Jiayuan Li (T) Darran Kavanagh (B) Eoin Conway (B)	Jonathan Hanley (T) Sahar El-Habbash (T) Orla Cahill (B)	Grainne Mc Kittrick (T) Quan Zhou (T)
GROUP I	**GROUP J**		
Michael Kelly (T) Ita Rowsome (B) Shi Jie (B) Loic Hervot (B)	James O'Sullivan (T) Mayowa Ajiboye (T) Kevin O'Brien (B) Yuanyuan Shen (B)		

- Do not make the colour as the *only* distinguishing information
- Notice what colours were used…

25

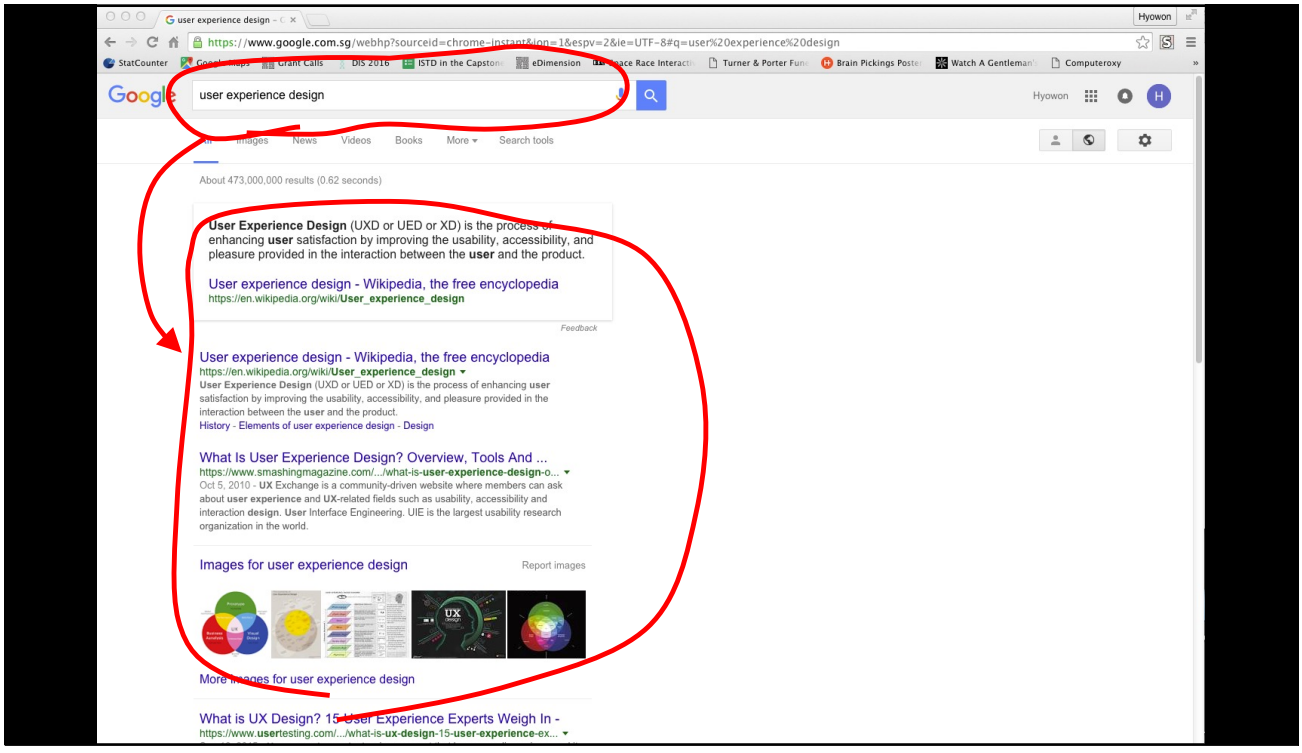


26

Layout

- Sequence of action

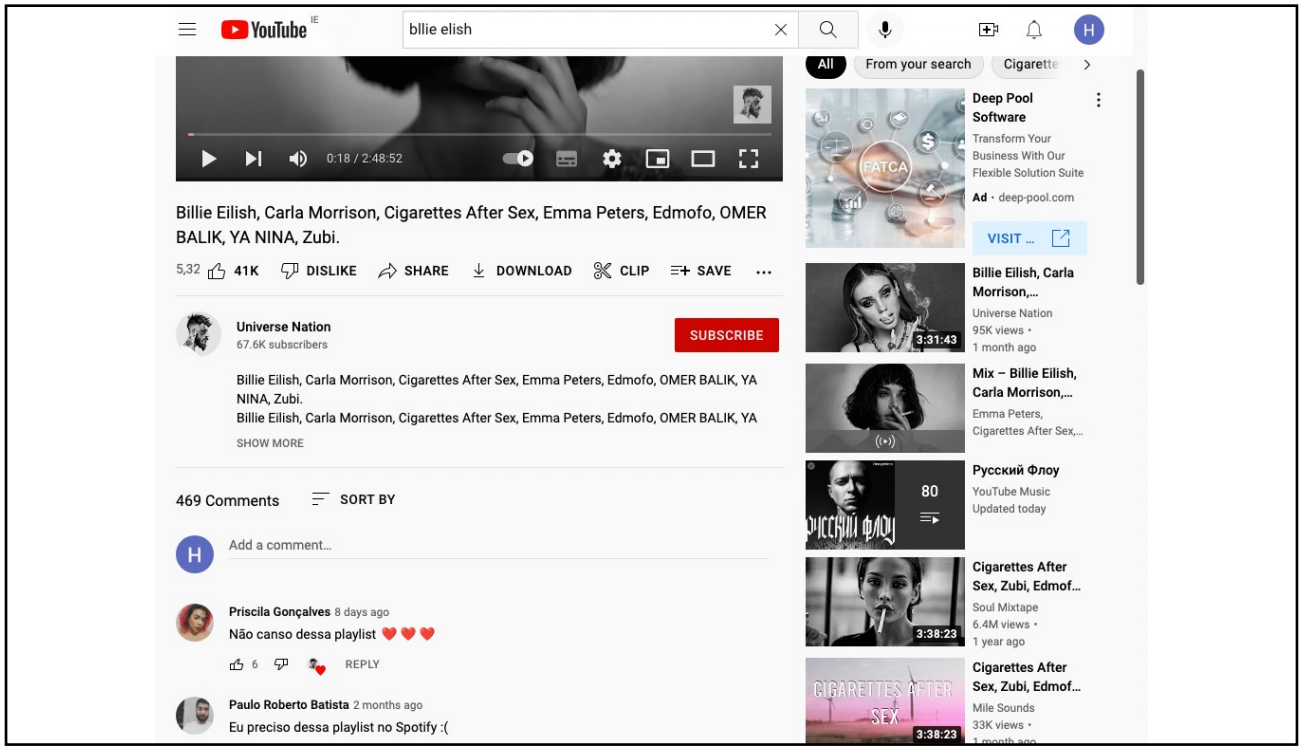
27



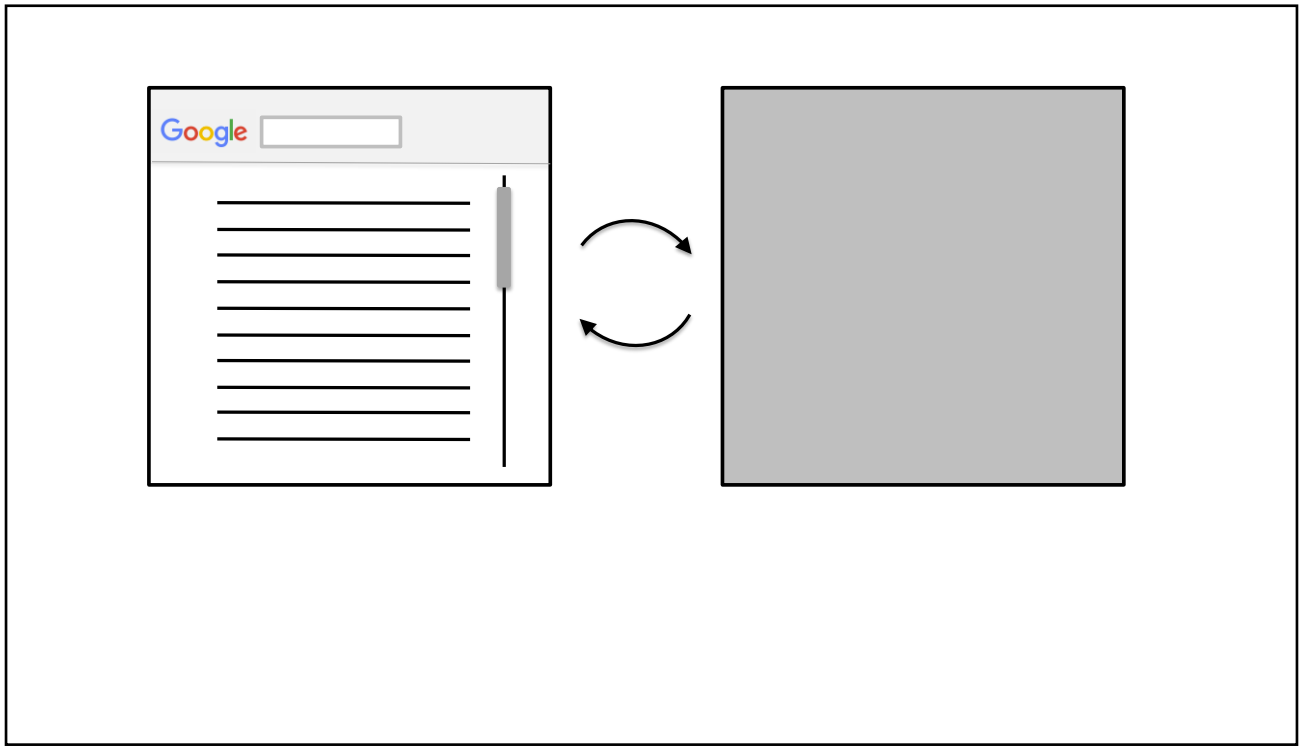
28


Google

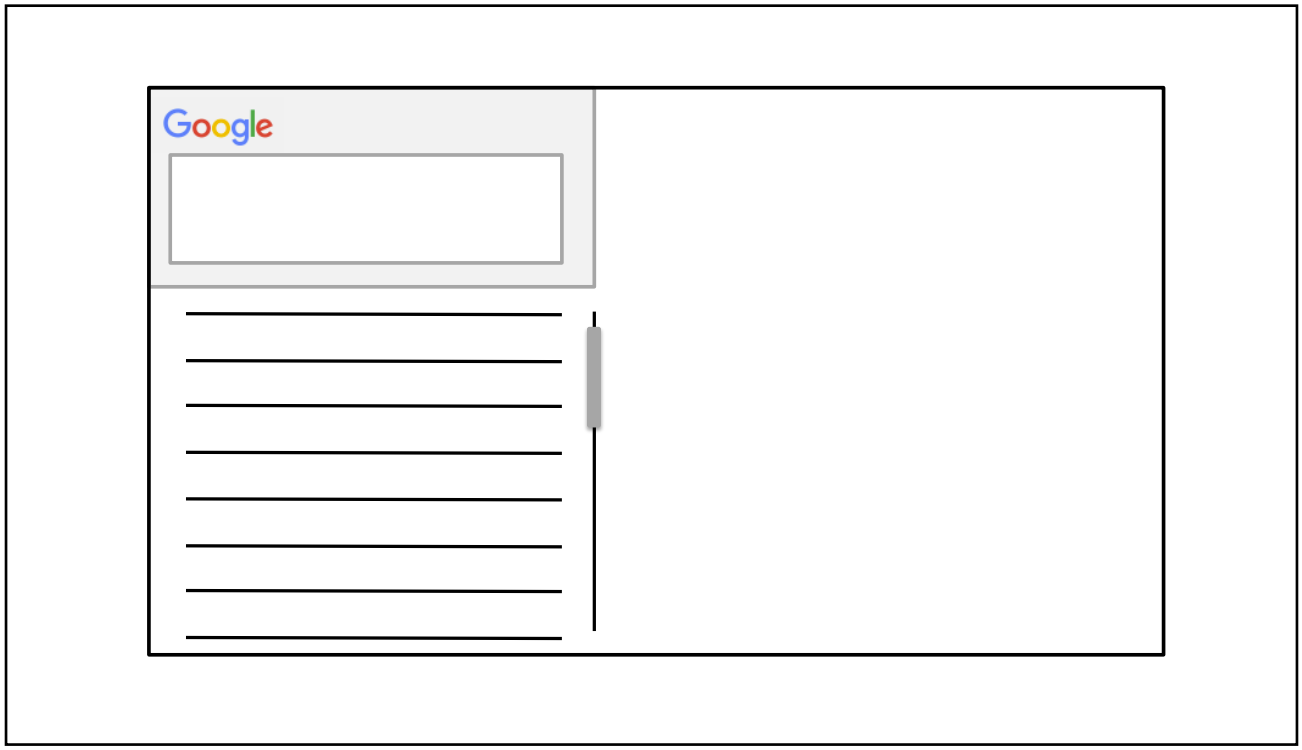
29


YouTube
bllie elish
All
From your search
0:18 / 2:48:52
Billie Eilish, Carla Morrison, Cigarettes After Sex, Emma Peters, Edmofo, OMER BALIK, YA NINA, Zubi.
5,32
41K
DISLIKE
SHARE
DOWNLOAD
CLIP
SAVE
Universe Nation
67.6K subscribers
SUBSCRIBE
Billie Eilish, Carla Morrison, Cigarettes After Sex, Emma Peters, Edmofo, OMER BALIK, YA NINA, Zubi.
Billie Eilish, Carla Morrison, Cigarettes After Sex, Emma Peters, Edmofo, OMER BALIK, YA
SHOW MORE
469 Comments
SORT BY
Add a comment...
Priscila Gonçalves 8 days ago
Não canso dessa playlist
REPLY
Paulo Roberto Batista 2 months ago
Eu preciso dessa playlist no Spotify :(
Deep Pool Software
Transform Your Business With Our Flexible Solution Suite
Ad · deep-pool.com
VISIT ...
3:31:43
Billie Eilish, Carla Morrison,...
Universe Nation
95K views
1 month ago
Mix – Billie Eilish, Carla Morrison,...
Emma Peters, Cigarettes After Sex,...
Русский Флоу
YouTube Music
Updated today
3:38:23
Cigarettes After Sex, Zubi, Edmof...
Soul Mixtape
6.4M views
1 year ago
Cigarettes After Sex, Zubi, Edmof...
Mile Sounds
33K views
3:38:23

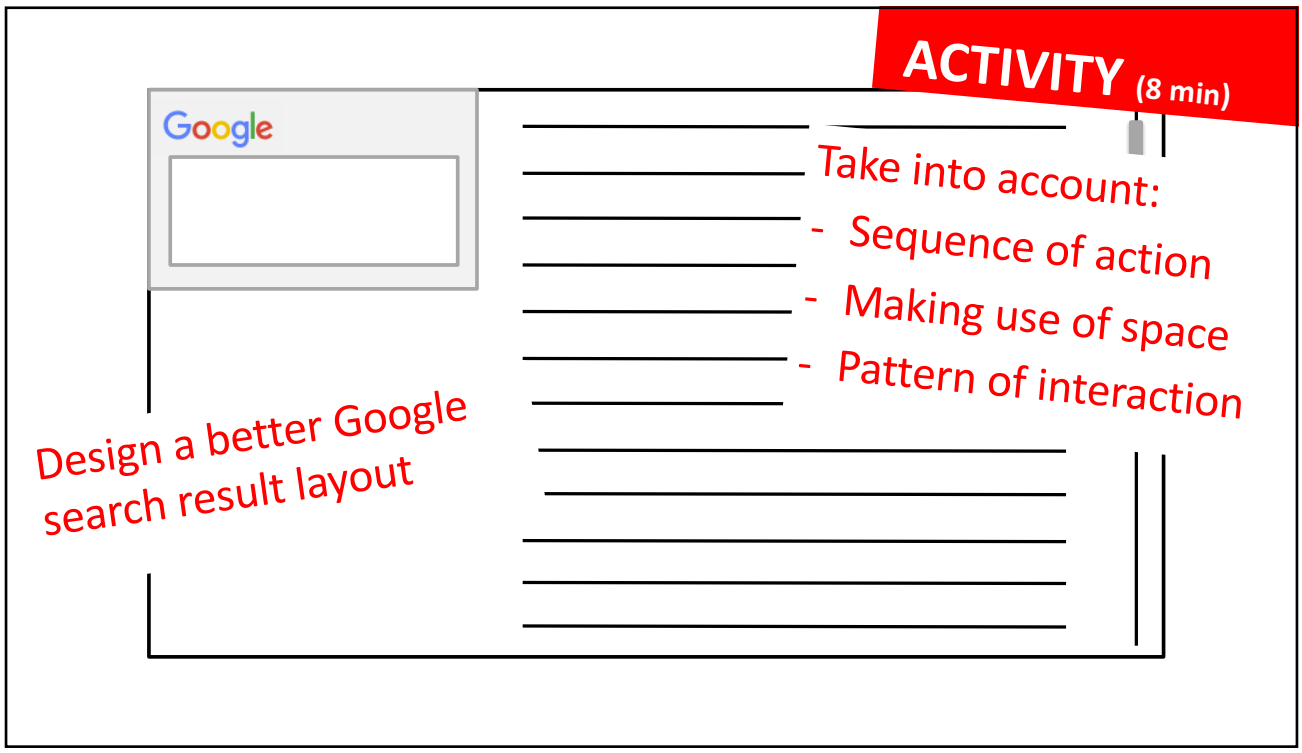
30



31



32



33

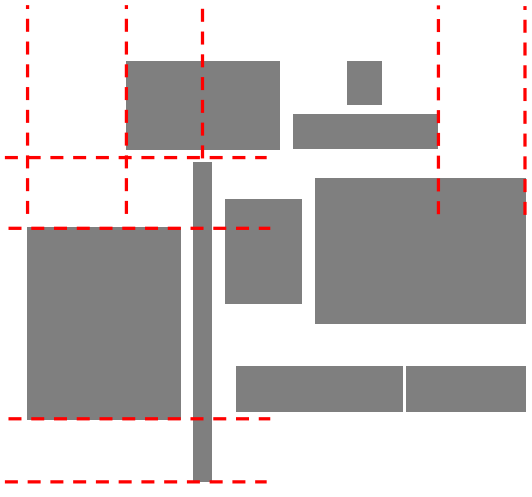
Layout

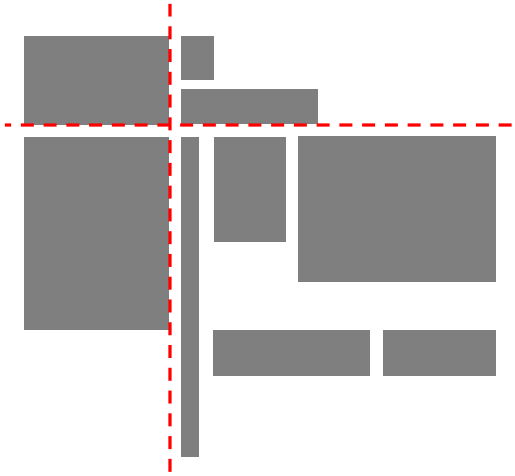
- Sequence of action
- Alignment

34

Align everything where possible

35





(Don’t centre the items – resulting in mis-alignment among everything)

36

Mr. and Mrs. Dursley, of number four, Privet Drive, were proud to say that they were perfectly normal, thank you very muc were the last people you'd expect to be involved in an strange or mysterious, because they just didn't hold wi nonsense.

Don't centre the main text in documents, including web/mobile UI

Mr. Dursley was the director of a firm called Grunnings, which made drills. He was a big, beefy man with hardly any neck, although he did have a very large mustache. Mrs. Dursley was thin and blonde and had nearly twice the usual amount of neck, which came in very useful as she spent so much of her time craning over garden fences, spying on the neighbors. The Dursleys had a small son called Dudley and in their opinion there was no finer boy anywhere.

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from Harry Potter and the Sorcerer's Stone, Chapter I

DCU

37

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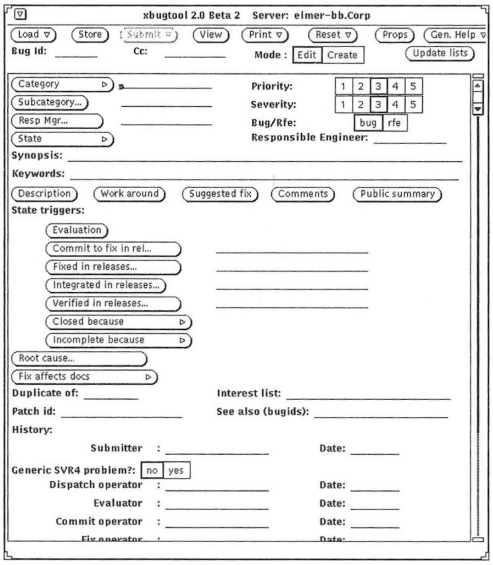
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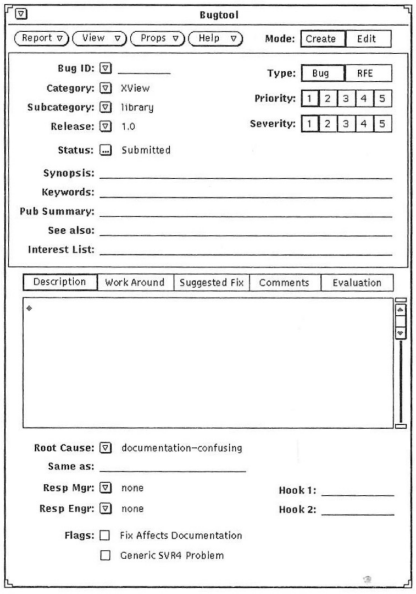
from Harry Potter and the Sorcerer's Stone, Chapter I

DCU

38

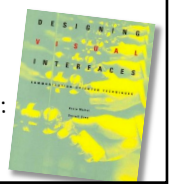


(a)

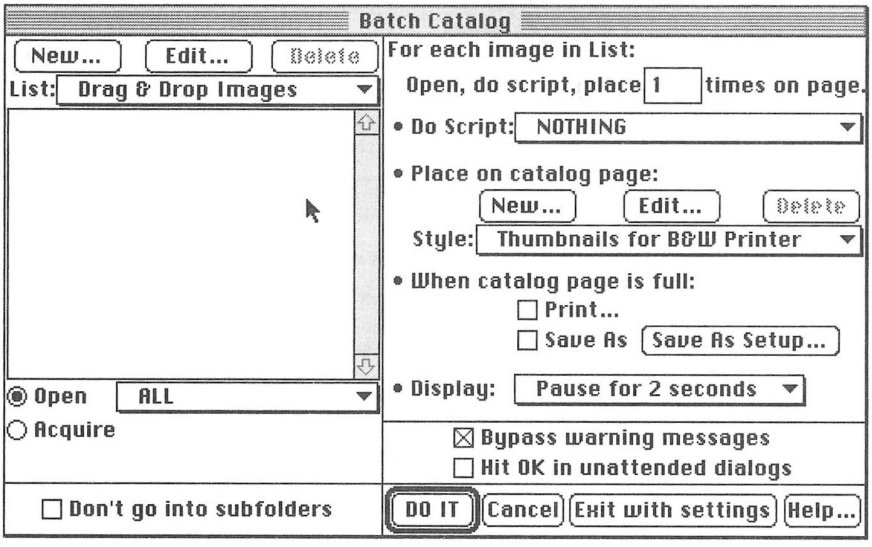


(b)

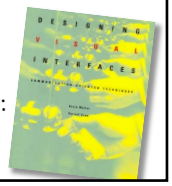
Kevin Mullet & Darrell Sano, Designing Visual Interfaces: Communication Orirented Techniques, 1995.



39



Kevin Mullet & Darrell Sano, Designing Visual Interfaces: Communication Orirented Techniques, 1995.

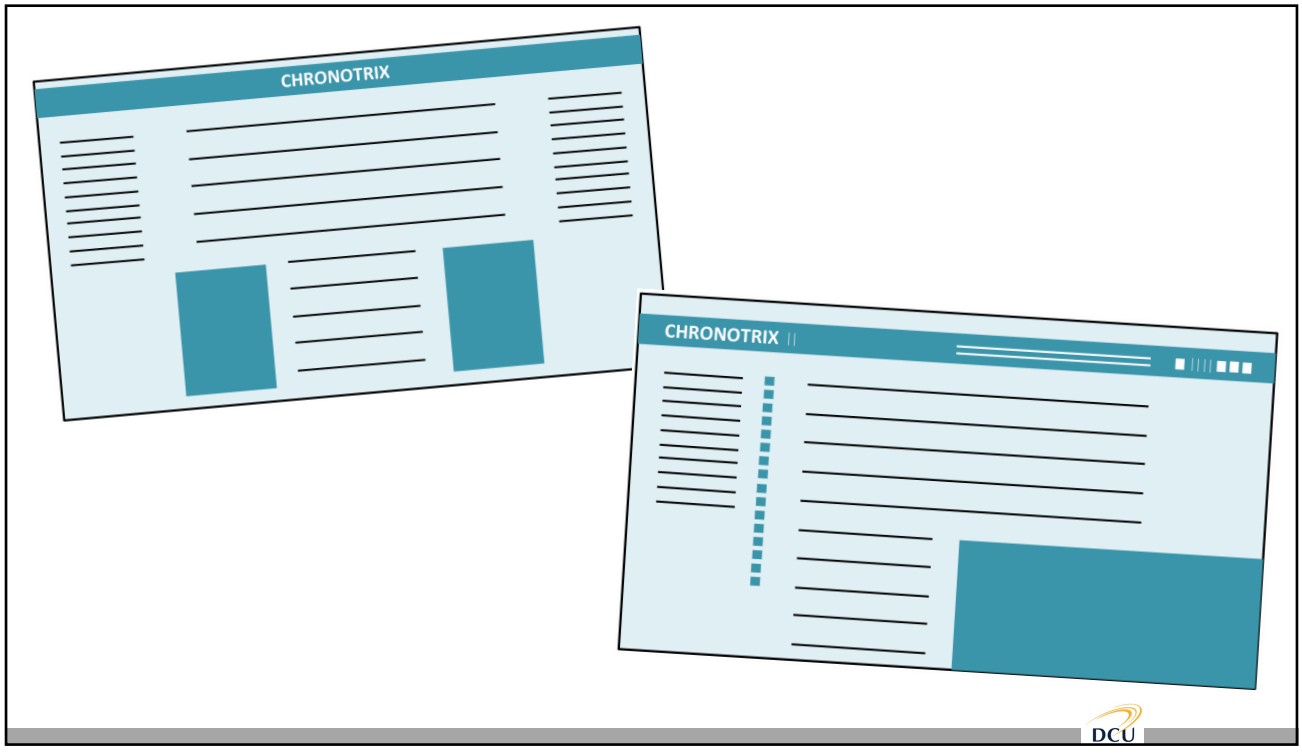


40

Layout

- Sequence of action
- Alignment
- Symmetry vs. Balance

41



42



43

Guideline 2:
Simple and natural dialogue

- Minimise the *artefacts*
- Grouping – Separating - Prioritising - Sharpening
- Readable fonts
- Good use of colour
- Layout concerns
- Emphasis

DCU

44


If everything is emphasized, the result is that nothing is emphasised
~Rio Mesa High School~
Spartans!
Peer Resources
Peer Helpers' Club
Pres.-Miguel Hernandez, Jevon Plunkett
V. P.-
Secretary-Amanda Gomez
Treasurer-
We Will Never Forget
SEPTEMBER 11, 2001
Peer Education--10/8-11
Peer Helper Training--10/11-12
Red Ribbon Week:--
Museum of Tolerance--
Fund Raising--
Happiness Day--
Diversity Week:--
US History 3CP
CLASS INFORMATON
Assignments:
Chap 1 Sect 4 Due/Quiz 10/5
TEST ALERT!!! 10/9
All of Chapter 1 + notes
& class activities
US History 3CPH
CLASS INFORMATION
AP --2001Exam stuff
DBQ Criteria
Colonies/D of I
Assignments:
Chap 1 Sect 4 Due/Quiz 10/5
TEST ALERT!!! 10/9
All of Chapter 1 + notes

45